Backtest referee

How far to trust the best backtest out of many tries

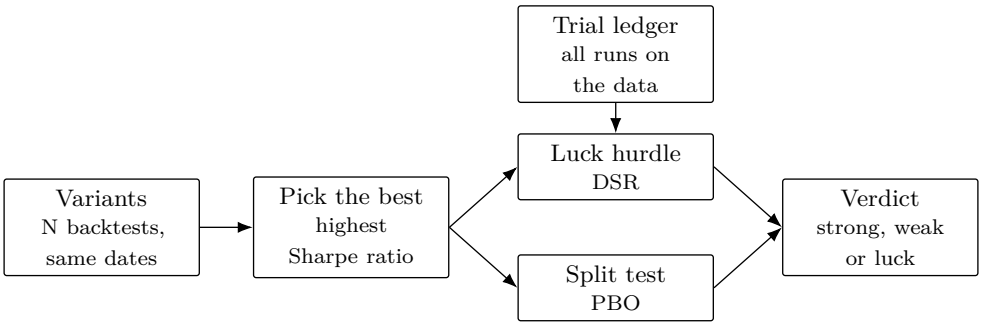
Universe	Strategy variants backtested on the same data; first used on 256 trend rules on SPY
Output	Verdict with deflated Sharpe ratio (DSR) and probability of backtest overfitting (PBO)
Version	0.1, October 2026
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Full spec	`PLAN.md` section 6.5 and `quantlab/validation.py`; no methodology document yet

Decision question

Is the best backtest from a search over many variants good enough to act on, once the search itself is taken into account? The referee does *not* forecast returns. It compares strategies with cash; the comparison with buy and hold is a separate step.

1 Intuition

The more variants someone tries, the better the best one looks by luck alone. In simulations for this project, the best of seven strategies with no skill at all shows a Sharpe ratio of about 1 over two years. The referee estimates how good the best of the tried variants would look if none of them had any skill, and asks how likely the winner is to be truly above that luck hurdle. A second check cuts the history into halves in 12'870 different ways and counts how often the winner of one half ends up in the bottom half of the other. Every variant ever run on the same data is counted in a permanent ledger, so repeated searches raise the hurdle.



2 Inputs

Input	Identifier	Used for
Daily excess returns of all variants	`backtest.run_many`	hurdle, DSR, PBO
The picked variant	highest full sample Sharpe ratio	DSR, holdout
Variants ever tried on this data	`output/trial_ledger.csv`	hurdle
Prices and cash rate (example)	Yahoo `SPY`; Ken French T bill	the backtests

3 How the score is built

$$\mathrm{DSR} = \Phi\left(\frac{(\widehat{SR} - SR_0)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\,\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\,\widehat{SR}^2}}\right)$$

where $\widehat{SR}$ is the winner's daily Sharpe ratio over T days, SR_0 the luck hurdle, $\hat{\gamma}_3$ and $\hat{\gamma}_4$ measure how lopsided and how fat tailed its daily returns are, and Φ turns the result into a probability. In words: the DSR is the probability that the winner's true Sharpe ratio lies above the hurdle. The hurdle grows with the number of variants tried and with how widely their results spread. Longer histories make the estimate more precise; large losses on single days make it less precise. PBO is the share of the 12'870 history splits in which the first half's winner lands at or below the middle of the second half. The verdict combines both numbers with thresholds fixed before any result is seen; *luck* below stands for *not distinguishable from luck*.

4 How to read the output

Verdict	Condition	Meaning
Strong	DSR at least 0.95 and PBO at most 0.20	The winner is very likely above the luck hurdle, and picking winners in this family works out of sample.
Weak	DSR at least 0.80 and PBO at most 0.50	Some evidence for the winner; treat its backtest Sharpe ratio as an upper bound.
Luck	anything else	The backtest is about as good as the best useless variant would look. Do not act on the specific rule.

A low PBO is no evidence for the winner by itself. PBO shows whether choosing within the family helps; DSR shows whether the chosen rule beats luck. In the first application PBO was 0.10 and DSR 0.58.

5 Worked example

256 trend timing rules on SPY, 11 April 1994 to 9 October 2026 (8'180 trading days), one day delay and 5 bp costs per trade. The best rule (50 day average above 250 day average, invested or in cash, scaled to 15% volatility, weekly) has a Sharpe ratio of 0.68; buy and hold has 0.53. The luck hurdle for 256 variants is 0.65, so DSR is 0.58, below 0.80; PBO is 0.10; verdict: not distinguishable from luck. Picked on 1994 to 2015 only, the best rule earned a Sharpe ratio of 0.65 afterwards against 0.75 for buy and hold. In words: the best trend rule looks about as good as the best of 256 useless rules would, and it fell behind buy and hold after the selection period.

6 Limitations

- Similar variants count as fewer independent tries, and the right count is uncertain: in the example the verdict is strong if the 256 rules count as 2.5 independent tries and not distinguishable from luck at 105 or 256. The referee uses the raw count
- The hurdle treats every difference between variants as luck, including real ones such as invested or in cash against long or short, which makes it strict
- Strategies are measured against cash: a rule that holds shares most of the time clears a zero hurdle easily, so compare with buy and hold separately
- A single PBO figure scatters widely: for pure noise it ranged from 0.16 to 0.84 across 20 simulated histories
- Only variants recorded in the ledger count. Ideas tried outside this code on the same data are missing, so the hurdle is a lower bound